SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTH207
Linear Algebra

Class Test 01

July 2026 Presentation

CLASS TEST 01 (1 hour)

This assignment is worth 15% of the final mark for MTH207 Linear Algebra.

Note to Students:

1. This class test contains **THREE (03)** questions.
2. This is an **OPEN BOOK** test. **Internet access is not permitted during the class test.**
3. Answer all questions on the blank pages provided.
4. Marks will only be awarded if FULL working is shown.
5. Before you submit your test, write your name and PI number on the cover page.

Student Name: _____

PI Number: _____

Question 1

Consider the linear system:

$$\begin{aligned} x+y+5z &= 5 \\ x+2y+3z &= 4 \\ x-y+a^2z &= a \end{aligned}$$

where $a \in \mathbb{R}$. Solve the system of linear equations to determine the conditions on a such that the linear system

(a) has no solutions.

(20 marks)

(b) has a unique solution

(10 marks)

(c) has infinitely many solutions

(10 marks)

Question 2

Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}.$$

Calculate the determinant of A and the inverse of A.

(30 marks)

Question 3

(a) Define the set $S := \{(0, a-\sqrt{7}b, b, 0) \in \mathbb{R}^4 \,|\, a, b \in \mathbb{R}\}$. Show that S is a vector subspace of $\mathbb{R}^4$ with respect to the usual vector addition and scalar multiplication in $\mathbb{R}^4$. Determine a basis for S.

(20 marks)

(b) Let

$$B = \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Determine whether the rows of B are linearly independent.

(10 marks)

---- END OF CLASS TEST ----